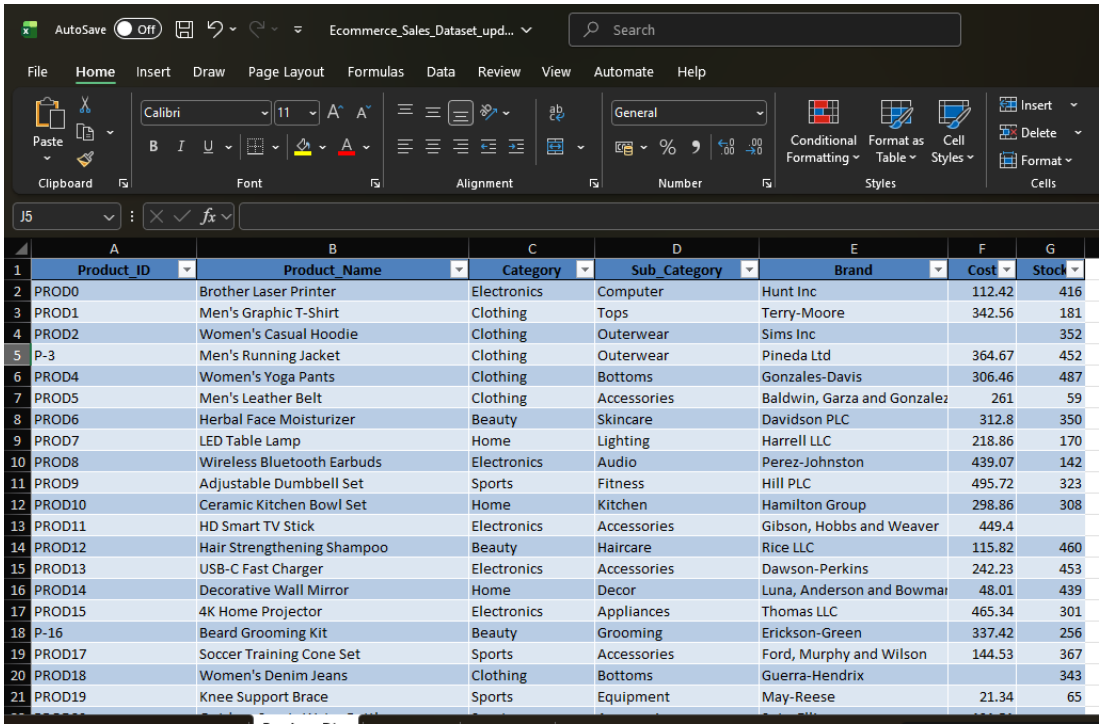
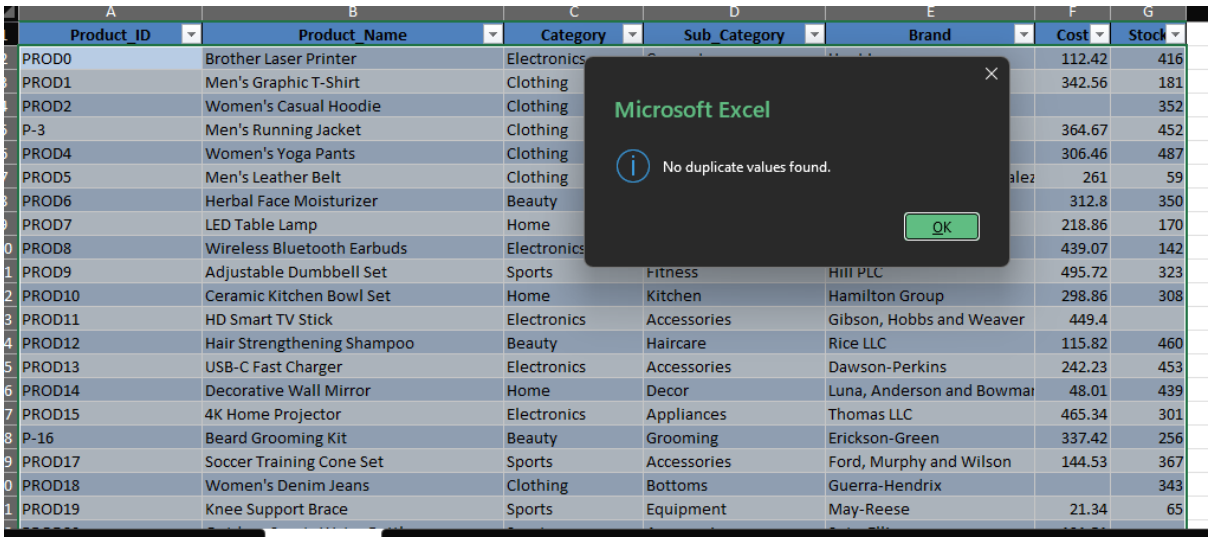
Data entry, import, and basic organization using tables



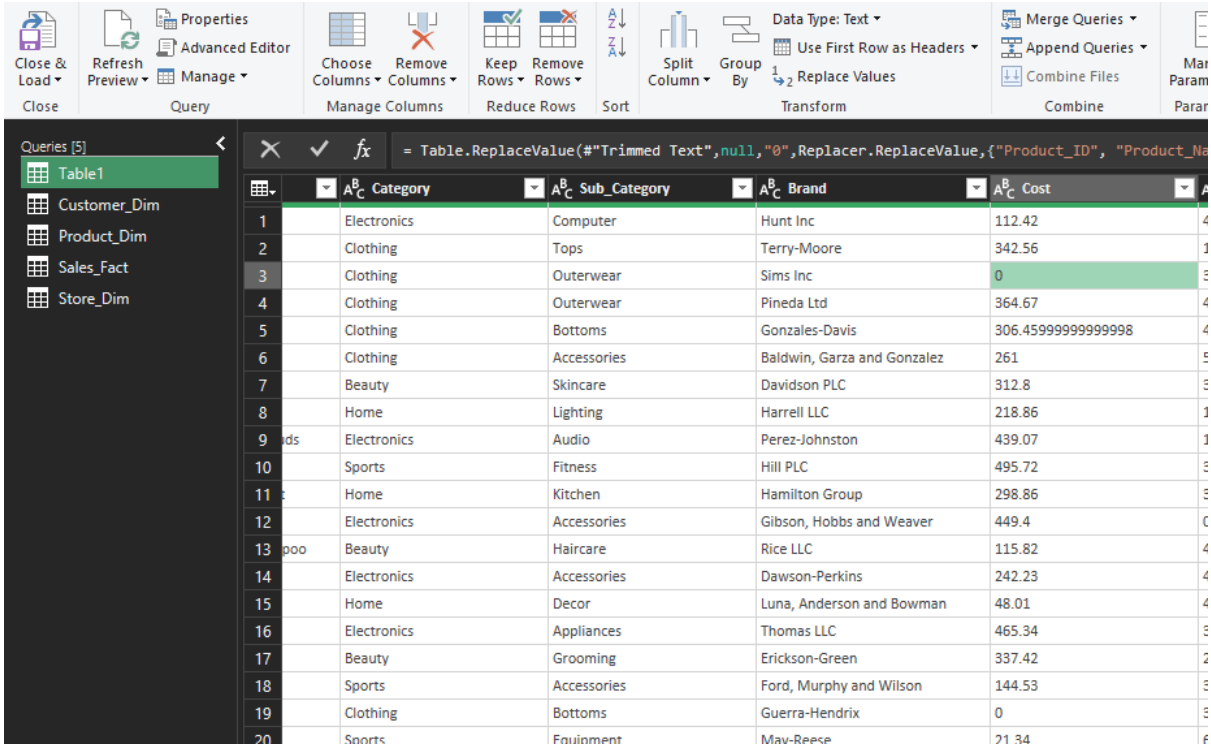
	A	B	C	D	E	F	G
1	Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock
2	PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	416
3	PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	181
4	PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc		352
5	P-3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	452
6	PROD4	Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306.46	487
7	PROD5	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	261	59
8	PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	312.8	350
9	PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	218.86	170
10	PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	142
11	PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	495.72	323
12	PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	308
13	PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	449.4	
14	PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	460
15	PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	453
16	PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson and Bowmar	48.01	439
17	PROD15	4K Home Projector	Electronics	Appliances	Thomas LLC	465.34	301
18	P-16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256
19	PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy and Wilson	144.53	367
20	PROD18	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix		343
21	PROD19	Knee Support Brace	Sports	Equipment	May-Reese	21.34	65

DATA CLEANING:
find/remove duplicates



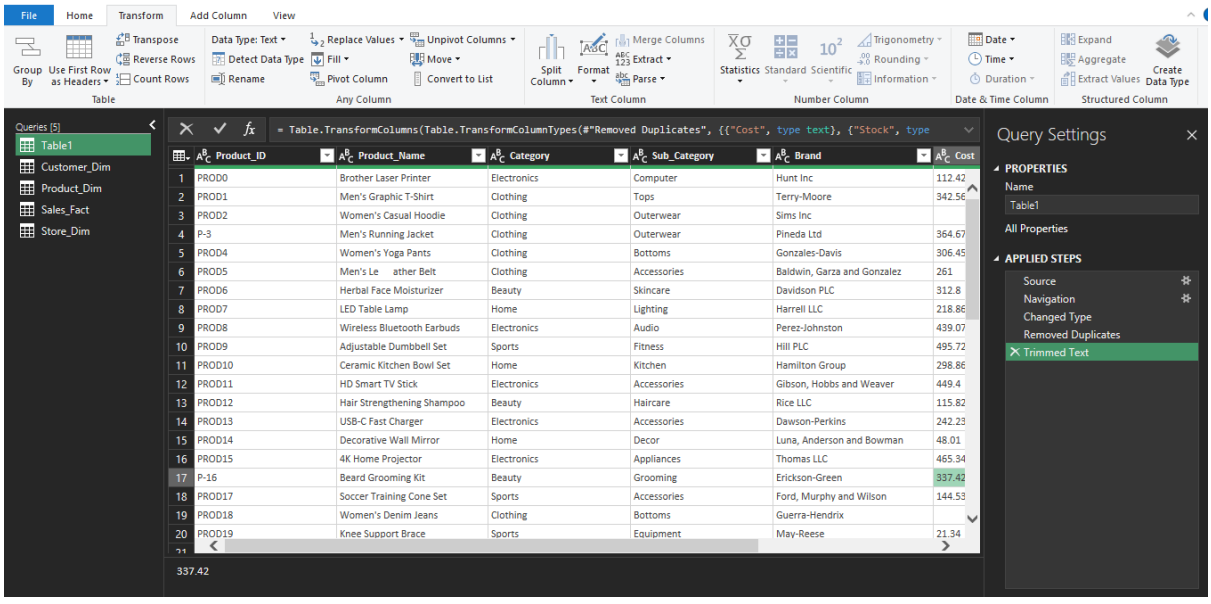
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21	PROD19	Knee Support Brace	Sports	Equipment	May-Reese	21.34	65

Handle missing



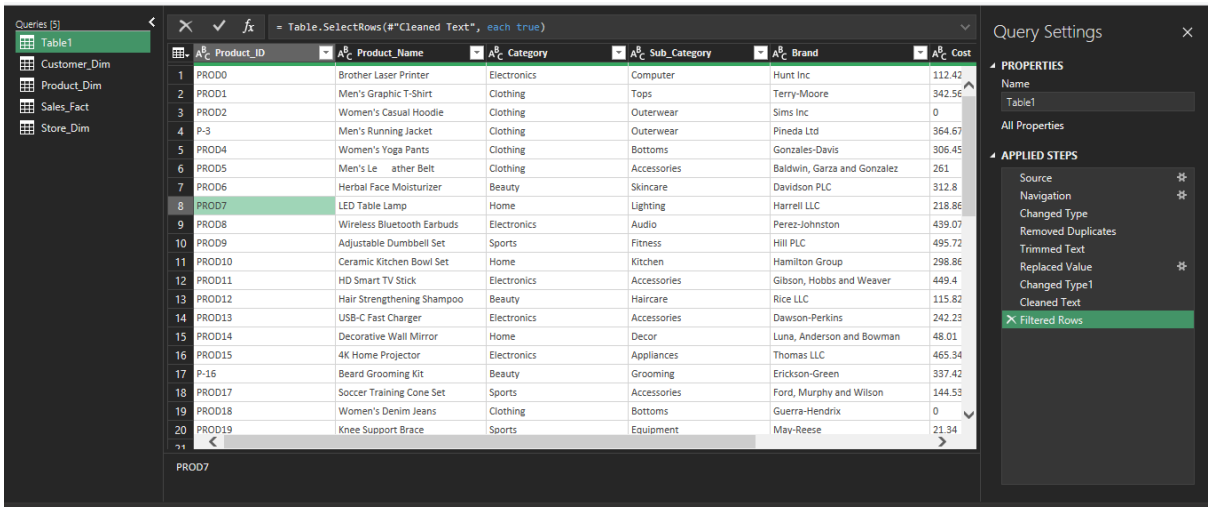
Missing value I replace the “0” in cost and stock column.

Inconsistent data



Inconsistent data I use trim function of all the columns.

correct formatting errors



First I do change data type for all columns.

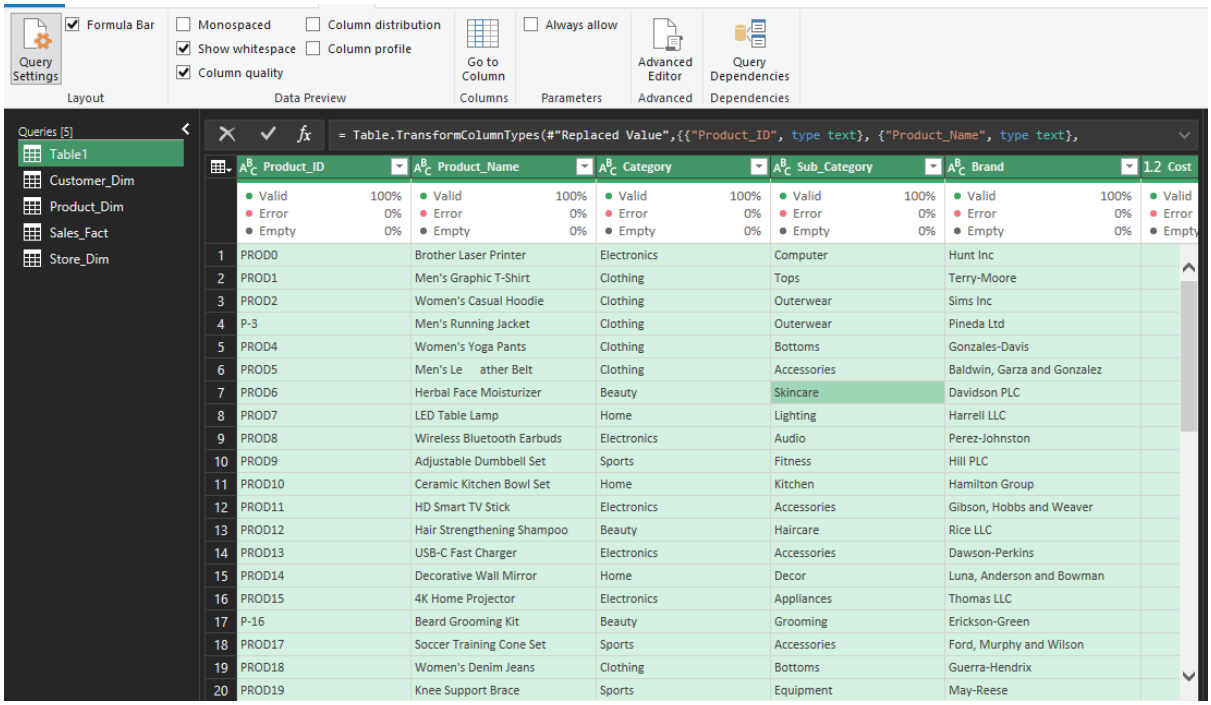
Then,

Text datatype I choose the capitalize each word format.

Number datatype I choose the whole number format.

Decimal datatype I choose the decimal number format.

Validate data



I use column quality column distribution, data type checks, and duplicate detection to identify missing values, errors, and duplicate records.

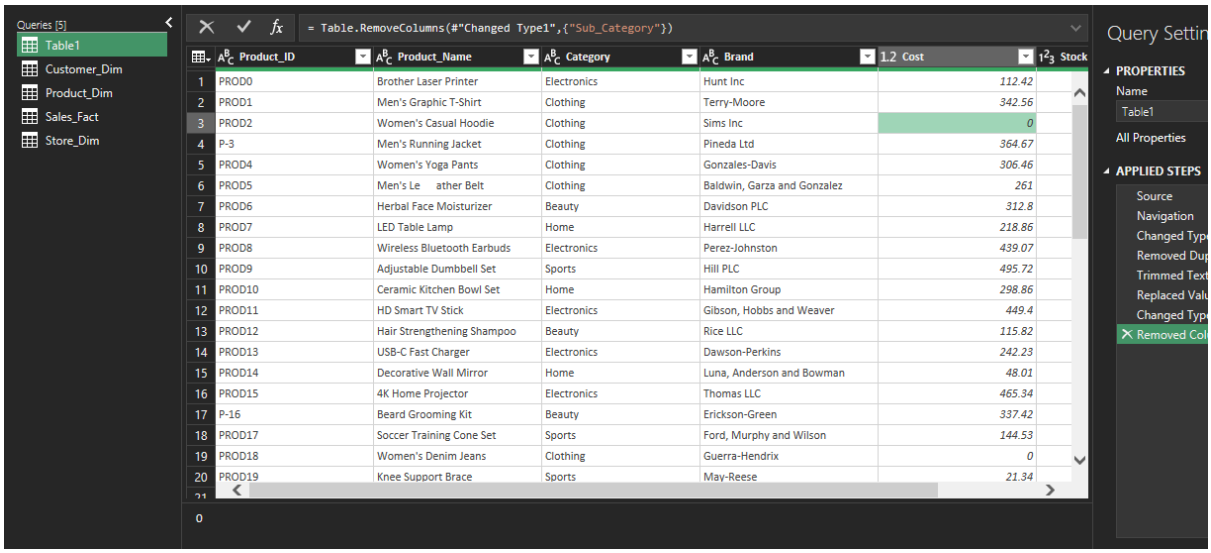
Go to view => column quality.

Remove the column that contains irrelevant data if it exists

I have 7 columns that are:

- Product id
- Product name
- Category
- Sub category
- Brand
- Cost
- Stock

Which column are unnecessary for the project requirements so I decided to the sub category column to remove based on a project that analyse the product performance.



Application of Excel Formulas for Calculations, Summaries, and Logical/Text Operations

1. Calculation Formulas

These formulas perform basic mathematical operations on data.

Formula	Purpose	Example
=SUM()	Calculates the total	Total price
=AVERAGE()	Calculates the average	Average price
=MIN()	Finds the minimum value	Lowest price
=MAX()	Finds the maximum value	Highest price
=COUNT()	Counts numeric cells	Number of prices

2. Summary Formulas

These formulas help summarize and analyze data.

Formula	Purpose
=COUNTA()	Counts non-empty cells
=COUNTIF()	Counts products in Accessories
=SUMIF()	Adds prices for Electronics
=AVERAGEIF()	Calculates the average price of Electronics

3. Logical Formulas

Logical formulas check conditions and return results.

Formula	Purpose
=IF()	Classifies prices
=AND()	Checks whether both conditions are true
=OR()	Checks whether either condition is true
=IFERROR()	Returns 0 if the calculation produces an error

4. Text Formulas

Text formulas help extract, combine, and clean text.

Formula	Purpose
=LEFT()	Extracts the first 3 characters
=RIGHT()	Extracts the last 3 characters
=LEN()	Counts characters
=TRIM()	Removes extra spaces
=UPPER()	Converts text to uppercase
=LOWER()	Converts text to lowercase
=CONCAT()	Combines text from two cells

Descriptive Statistics in Excel

1. Excel formulas

Statistic	Excel formula	Meaning
Mean	=AVERAGE(F2:F101)	Average sales
Median	=MEDIAN(F2:F101)	Middle value
Mode	=MODE.SNGL(F2:F101)	Most frequently occurring value
Standard deviation	=STDEV.S(F2:F101)	How much values vary
Variance	=VAR.S(F2:F101)	Measure of data spread
Minimum	=MIN(F2:F101)	Lowest sales value
Maximum	=MAX(F2:F101)	Highest sales value
Count	=COUNT(F2:F101)	Number of numeric values
Sum	=SUM(F2:F101)	Total sales
Range	=MAX(F2:F101)-MIN(F2:F101)	Difference between maximum and minimum

Step apply:

Step 1: Enable Analysis ToolPak
File → Options → Add-ins → Manage Excel Add-ins → Go
Analysis ToolPak checkbox select panni OK click pannunga.

Step 2: Open Data Analysis
Data tab click pannunga. Right side-la Data Analysis select pannunga.

Step 3: Select Descriptive Statistics
Input Range: Sales_Fact!J1:J2001
Grouped By: Columns

Data visualization:

create at least three chart types, build an interactive dashboard, and use basic PivotCharts

